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Faculty Information and Communication Technology Department of Computer Science

		PROJECT PROPOSAL	√	**Total Marks**	**80**
SUBJECT CODE:		SFG117V / ISE117V			
Examiners		Dr. HD Masethe			
Submission DATE: (For Office use only)		11 -April 2026@23:59			
Group Name		Royal Hackers			

Instructions

- Group yourself in groups of 3-6 students
- Project may be a web, mobile, desktop, cloud, distributed, service-oriented, data-driven and AI-enabled system with a proper user interface.
- The proposal must clearly explain the AI purpose, data needs, risks, evaluation approach, and responsible-use controls
- The proposal should show clear alignment to the module learning outcomes and realistic semester-level scope.
- Teams must consist of 3–6 students. Define roles clearly to promote accountability, collaboration, and workload balance. A student may play more than one role, but responsibilities must be explicit.

Student Name and Initials	**Student Number**	**Proposed role**	**Key responsibilities**
1. Basopu A	230172234	Product Owner	DevOps
2. Nkambule MM	208254049	Developer	Backend
3. Zenani K	224634773	Developer	Backend
4. Madela N	218729797	Scrum master	Documentation
5. Mtsweni SH	223019510	Developer	Frontend
6. Nkosi S.A	207060550	Developer	Frontend

1.Project Overview

Complete this section with a concise, persuasive summary of the project idea. The overview should explain the problem, intended users, proposed solution, expected impact, and technical scope.

Project Title: EduTrack SA

Item	Template guidance for students
Problem statement	In South Africa, learners in disadvantaged schools (Quintiles 1-3) face significant challenges in achieving satisfactory results in the National Senior Certificate (NSC) examinations. According to the Department of Basic Education (2023), only 28,851 matriculants scored at least 60% in both mathematics and physical science, with mathematics participation declining from 53% in 2008 to 38.5% in 2023 (DBE, 2024). These challenges stem from limited access to learning resources (only 35% of Quintile 1-3 schools have computer laboratories), high mobile data costs (average 1GB costs approximately R34.28 as of 2025), unreliable electricity supply during peak hours, and a lack of effective digital engagement (over 60% of educators have only basic ICT training). Approximately 33% of South Africa's population resides in rural areas, where only 40% of households have reliable internet connectivity (Ken Research, 2025). As a result, learners struggle academically, reducing their chances of accessing higher education and employment opportunities.
Proposed solution	The proposed system is a dual-stakeholder digital academic monitoring and engagement platform that places equal emphasis on learner success and teacher empowerment. The system enables teachers to upload structured learning content and learners to access it and complete assessments. Artificial Intelligence (AI) is integrated to automatically generate quizzes from uploaded content using Large Language Models (LLMs) via API, reducing teacher workload by an estimated 30–40%. The system is built as a Progressive Web App (PWA) with offline-first capabilities using Service Workers and IndexedDB, enabling learners to access content without continuous connectivity. A data-saving mode reduces data usage by up to

	60% through text-only rendering, image compression, and lazy loading. Gamification elements (points and streaks) drive daily learner engagement.
Project objectives	• Improve learner engagement through gamification (points, streaks, leaderboards) • Empower learners to track their own progress through personalised dashboards • Provide accurate and automated assessment tools via AI-generated quizzes with teacher review • Enable teachers to monitor learner progress in real-time through teacher dashboards • Reduce teacher workload by 30–40% through AI-generated quizzes • Support early identification of struggling learners (scores below 50% for two or more consecutive quizzes) • Enable school management to make data-driven resource allocation decisions
Scope boundaries	**Included:** • Web-based PWA for learners and teachers • Content upload (.txt, .pdf, .docx; max 10MB) • AI-generated quizzes (5–10 MCQ or short-answer questions) • Learner dashboard – progress, points, streaks, subject breakdown • Teacher dashboard – class average, at-risk flagging, intervention tracking • School management dashboard – aggregate analytics • Offline-first core with Service Worker and IndexedDB • Data-saving mode, gamification, sync queue, email notifications **Excluded:** • Native mobile app (iOS/Android) • Integration with DBE/SIS systems • Custom AI model training (pre-trained LLM APIs only) • Parent/guardian portal (Phase 2) • Real-time video conferencing • Predictive analytics • Offline AI generation
Expected deliverables	• Requirements specification document • System design diagrams (UML, architecture) • Working prototype/system • Test cases and testing report

	• Final presentation and report
AI opportunity / AI role in the project	AI will be used to automatically generate quizzes from uploaded learning content using Large Language Models (OpenAI GPT-4o-mini primary; Groq Llama 3 fallback). The AI processes text-based content and produces multiple-choice and short-answer questions. All AI-generated content requires teacher review before publication to learners.

2.Learning Outcome Alignment Matrix

Show how the proposed project will address each module learning outcome. The explanation should be specific, practical, and linked to project activities or artefacts.

LO	Learning outcome	Project-specific evidence / alignment
A.1	Apply Software Process Improvement Techniques	Use sprint retrospectives to identify inefficiencies and improve workflows. Defect tracking and root-cause analysis will be used to reduce recurring issues
B.1	Understand Agile and DevOps in Modern Software Processes	Adapted Scrum (async daily updates, weekly sync), GitHub for version control, GitHub Actions for CI/CD, Docker for containerisation, Vercel for frontend deployment.
C.1	Understand Software Process Metrics and Quality Assurance	Test coverage (≥75%), defect density (<4/KLOC), test pass rate (≥90%), sprint velocity, and DRE tracked per sprint using SonarQube and GitHub Actions dashboards.
D.1	Understand Agile Methodologies and Frameworks	Adapted Scrum for distributed student team: evening sprint planning, async WhatsApp/Discord daily updates, weekly sync meetings, sprint reviews and retrospectives every two weeks.
E.1	Analyse and Specify Software Requirements	Structured requirements: functional, non-functional and AI-specific, with MoSCoW prioritisation and traceability matrix.
F.1	Apply Software Architecture and Design Patterns	Client-server architecture with MVC, Singleton (database connection), Repository (data access), and Circuit Breaker (LLM API failover) design patterns applied.
G.1	Develop and Execute Test Cases	Unit, integration, and system testing
H.1	Apply Techniques for Improving Software Dependability	Secure login and data protection
I.1	Apply Software Reuse Techniques and Strategies	Reuse of frameworks such as FastAPI, Next.js, and Supabase will reduce development time and improve reliability.

J.1	Analyse Distributed System Architectures	Cloud-based distributed deployment: frontend on Vercel, containerised FastAPI backend on Render/Azure Container Apps, Supabase cloud database. Client-server with distributed components.
K.1	Evaluate Real-World Service Oriented Architecture (SOA)	The system will use a service-oriented approach where the frontend, backend, and AI services operate independently and communicate via APIs
L.1	Analyse Emerging Trends in Software Engineering	The project incorporates AI-assisted learning, cloud-native deployment, and backend-as-a-service (BaaS), aligning with modern software engineering trends.

3.Problem Context and Justification

3.1 Domain Context

This project operates within the South African basic education domain, specifically targeting no-fee public schools in Quintiles 1–3. These schools serve learners from the most economically disadvantaged communities in urban townships and rural areas. The project aligns directly with the Department of Basic Education's strategic objectives for digital transformation, as outlined in the Action Plan 2024: Towards the Realisation of Schooling 2030 (DBE, 2020).

3.2 Current Pain Points and Evidence

Pain Point	Evidence / Statistic	Source
Limited computer lab access	Only 35% of Quintile 1–3 schools have computer laboratories. Full ICT rollout estimated at R30.6 billion.	DBE Action Plan (2020)
High mobile data costs	Average 1GB mobile data costs approximately R34.28 as of 2025.	MyBroadband (2025)
Declining STEM participation	Mathematics participation declined from 53% (2008) to 38.5% (2023).	DBE Progress Review (2024)

Pain Point	Evidence / Statistic	Source
Low NSC STEM achievement	Only 28,851 matriculants scored ≥60% in both Maths and Physical Science in 2023.	NSC Technical Report (2023)
Poor educator ICT skills	Over 60% of educators have only basic ICT training.	DBE Action Plan (2020)
Rural connectivity gap	33% of South Africans live in rural areas; only 40% have reliable internet.	Ken Research (2025)

3.3 Stakeholder Analysis

Stakeholder	Role	Primary Benefit
Learners	End users – access content, complete quizzes, track progress	Access offline; immediate quiz feedback; personalised dashboard showing strengths, weaknesses, and improvement trends; gamification engagement
Teachers	Content creators, quiz reviewers, progress monitors	Upload content once; AI generates quizzes; real-time class performance view; at-risk learner identification; reduced workload by 30–40%
School Management	Oversight and reporting	Aggregate analytics; school-wide performance trends; data-driven resource allocation decisions

3.4 Suitability for Advanced Diploma Level

This project is appropriate for the Advanced Diploma in Computer Science because it integrates multiple advanced paradigms: web development (Next.js PWA), backend API design (FastAPI), cloud database management (Supabase PostgreSQL), AI service integration (LLM APIs via prompt engineering), offline-first architecture (Service Worker, IndexedDB), and distributed cloud deployment (Vercel, Render). It requires demonstrable competencies in Agile project management, DevOps pipeline design, quantitative quality assurance, software dependability engineering, service-oriented architecture , and responsible AI integration. The project also addresses a genuine societal challenge — educational inequality in South African schools — providing real-world impact and motivation.

3.5 Assumptions and Constraints

Assumptions:

- Teachers have basic digital literacy to upload content.
- Learners have access to a smartphone or low-end device with a modern browser.
- Schools will provide teacher and learner consent for data collection.

Constraints:

- No integration with DBE or SIS systems in Phase 1.
- AI quiz generation requires internet connectivity.
- LLM API rate limits may affect concurrent uploads.

4.Requirements Analysis and Specification

This section addresses learning outcome E.1. The project proposal must show that the team can elicit, analyse, prioritise, and specify software requirements in a structured way. If the system includes AI features, requirements must also specify data inputs, human oversight, expected outputs, failure conditions, and measurable acceptance criteria for the AI behaviour.

Requirement area	Student response
Stakeholders	• Learners • Teachers • School management (Principal, Deputy Principal)
Elicitation approach	• Interviews with teachers and learners • Observation of classroom challenges • Use of user stories
Functional requirements	• User registration and login • Access learning content offline • Complete quizzes with auto-save • Automatic MCQ scoring • Gamification (points, streaks) • Content upload by teachers • AI quiz generation • Teacher review and edit AI questions • Teacher dashboard • School management dashboard
Non-functional requirements	• Page load time (< 1.5 seconds on 3G connection) • API response time (< 500ms for all API endpoints) • AI quiz generation time (< 30 seconds per request) • System uptime (≥ 99.5% per month (Vercel + Supabase SLA)) • JWT token expiry • Data encryption • Concurrent users
Prioritisation method	• Must have: Login, content upload, AI quiz generation, quiz taking, auto-scoring, secure authentication. • Should have: Learner dashboard, gamification, offline access, teacher dashboard, sync queue, performance targets. • Could have: School management dashboard, email notifications, data-saving mode, teacher review interface

	• Won’t have(Phase 1): Parent portal, DBE/SIS integration, native mobile app, predictive analytics, offline AI generation
Validation approach	Requirements will be validated through: • Stakeholder feedback (teachers review features) • Prototyping and demonstrations • User acceptance testing (UAT) • Review sessions with team members
AI-specific requirements (if applicable)	• Input: Text-based learning content • Output: Quiz questions • Accuracy: Questions must match content • Human oversight: Teachers can review quizzes • Fallback: Manual quiz creation if AI fails

5.Agile Process, DevOps, and Project Management Plan

This section should demonstrate learning outcomes A.1, B.1, and D.1. The team should propose how it will organise work using an Agile methodology and how DevOps practices will support integration, and feedback.

Element	Student Response
Agile framework	Adapted Scrum — modified for distributed student team with asynchronous communication. Sprints are 2 weeks each over a 10-week semester (5 sprints total).
Sprint cadence	2-week sprints. Sprint planning (Monday evening, 2 hours), mid-sprint weekly sync (Thursday evening, 1 hour), sprint review + retrospective (last day of sprint, 2 hours combined).
Async daily updates	Each member posts to the WhatsApp/Discord channel before 22:00 daily using the format: Yesterday: [completed task] Today: [planned task] Blockers: [none / description]
Agile artefacts	Product backlog (prioritised PBIs with story points), sprint backlog (tasks 4–16 hours each), GitHub Projects Kanban board (Backlog → In Progress → Review → Done), Definition of Done checklist, burndown chart (updated every 2–3 days), velocity tracking.
DevOps pipeline	• **Version Control:** GitHub with feature branch workflow; pull requests required for merges to main • **CI/CD:** GitHub Actions for automated unit/integration testing, linting, and deployment — no human intervention required for test execution • **Containerisation:** Docker for consistent backend environments (eliminates "works on my machine" issues) • **Frontend Deployment:** Vercel — preview deployments on PR, production on merge to main • **Backend Deployment:** Containerised FastAPI service on Render/Azure Container Apps • **Monitoring & Logging:** Prometheus + Grafana for metrics; ELK Stack (Elasticsearch, Logstash, Kibana) for centralised logging
Process improvement	Sprint retrospectives held at end of each 2-week sprint (1 hour). Metrics tracked: sprint velocity (story points

Element	Student Response
	completed), defect rates (defects per KLOC), and Defect Removal Efficiency (DRE = internal defects / total defects × 100%; target ≥ 85%). Actions tracked in GitHub Issues and reviewed at next sprint planning.
AI-assisted workflow	AI tools (GitHub Copilot) assist with code suggestions, documentation, and test case templates. All AI-generated code must be reviewed and tested by a developer before merging. No AI output merged without human review.

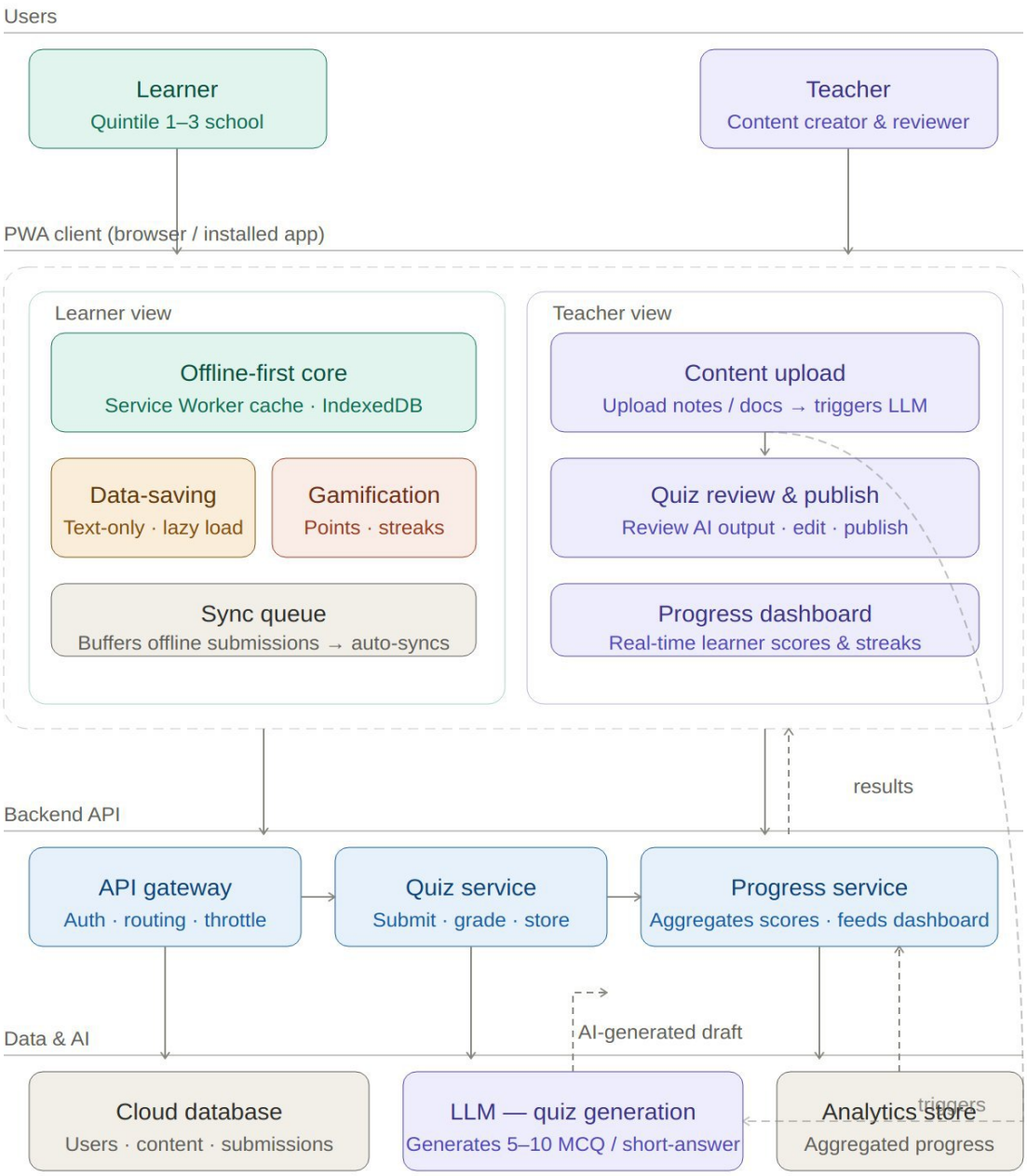
6.Software Architecture and Design

This section addresses learning outcomes F.1, J.1, and K.1. Students should show the proposed structure of the solution and justify architecture decisions.

Architecture decision area	Proposed decision	Rationale
Architecture style	Client-server architecture with cloud-based deployment	Supports scalability, centralized data management, and accessibility via web applications
Deployment model	Distributed web application hosted in the cloud	Ensures high availability and allows access from multiple devices and locations
Main components / services	• Frontend (User Interface) • Backend (API) • Database • AI module	Separation of concerns improves maintainability, scalability, and modular development
Design pattern(s)	• MVC (Model-View-Controller) • RESTful API design	MVC separates logic, UI, and data; REST APIs enable communication between components
Database / persistence choice	Supabase (PostgreSQL) will be used for data storage due to: • Scalability • Ease of integration with modern web frameworks	Provides scalability, real-time features, built-in authentication, and easy integration
Integration / API strategy	• REST APIs between frontend and backend • Backend communicates with AI services via API calls	Enables loose coupling between components and easy integration of external AI services
Service orientation (SOA)	Frontend, backend, and AI module operate as independent services communicating via RESTful APIs (SOA principles)	Loose coupling; each service can be updated, scaled, or replaced independently; aligns with LO K.1

Architecture trade-offs	Supabase creates vendor dependency; LLM API latency adds 5–25 seconds to quiz generation; cloud costs scale with usage	Mitigations: Supabase open-source self-hosting option; circuit breaker + fallback for AI; free tiers within semester scope
AI component placement	AI module sits at backend layer: teacher uploads content via frontend → backend extracts text → sends to LLM API → returns questions → teacher reviews via frontend	AI is decoupled from core app; failure of AI service does not prevent system operation; fallback to manual quiz available

System Over View Diagram



7.Software Reuse, AI Integration, and Emerging Technologies

This section addresses learning outcomes I.1 and L.1 while also requiring students to explain how AI will be used, evaluated, or deliberately excluded from the project. Teams must show how they will leverage existing assets responsibly, evaluate emerging technologies critically, and justify any use of AI tools, models, or intelligent features from both engineering and ethical perspectives.

Area	Student response
Reuse strategy	• FastAPI for backend development • Next.js for frontend • Supabase for database and authentication • Pre-trained LLM APIs for quiz generation
Benefits of reuse	**Benefits of Reuse** • Reduces development time • Improves reliability • Minimises bugs • Allows focus on core features
Risks of reuse	**Risks of Reuse** • Dependency on third-party services • Possible security vulnerabilities • API limitations or costs • Vendor lock-in
Emerging trend(s)	• AI-assisted education systems • Cloud-native development • Backend-as-a-Service (BaaS)
Relevance to project	These technologies enable rapid development, scalability, and integration with AI services, making them highly suitable for this system.
AI use case or justification for no AI	AI is used to automatically generate quizzes from educational content, improving efficiency and reducing teacher workload.
AI method / tool choice	• Large Language Models via API: OpenAI GPT-4o-mini (primary, cost-efficient); Groq Llama 3 (fallback, open-source) • Prompt engineering to generate: ○ Multiple-choice questions

	○ Short-answer questions
Data needs for AI	• Input: Teacher-uploaded educational content • Output: Quiz questions • No external dataset required (reduces privacy risk)
Responsible AI and limitations	• 100% teacher review before publication (human oversight mandatory) • AI outputs may contain errors → fallback to manual editing • Ensure no sensitive data is shared with AI services
AI evaluation plan	• Accuracy: teacher relevance rating ≥4/5 for 85% of questions (measured in UAT with 5 teachers) • Relevance to content: validated against uploaded material per quiz • Error rate: <5% of API calls return unusable output • Latency: AI response <30 seconds; teacher satisfaction: post-UAT survey score ≥4/5

8.Quality Assurance, Metrics, and Testing Strategy

This section addresses learning outcomes C.1 and G.1. Students must define how they will measure quality and how they will verify the system using structured testing. Where AI is used, the proposal should also explain how the team will test prompt behaviour, model outputs, fairness, robustness, and failure handling.

Area	Template guidance
Quality assurance activities	• Code reviews • Pair programming • Automated testing • Continuous integration checks
Software metrics	• • Defect density (target: < 4 defects/KLOC) • Test pass rate (target: ≥ 90%) • Code coverage (target: ≥ 75%) • Sprint velocity (story points completed per sprint)
Testing levels	• Unit testing • Integration testing • System testing • User acceptance testing
Test case design approach	• Functional requirements • User stories • Edge cases (e.g., invalid inputs)
Acceptance criteria	• System performs all required functions • No critical bugs • AI-generated quizzes are relevant • System is user-friendly
AI quality metrics (if applicable)	• Accuracy: ≥85% of questions rated relevant by teacher (UAT) • Hallucination / error rate: <5% unusable outputs per session • Response latency: <30 seconds per generation request

	• Bias checks: teacher review checklist completed for 100% of published quizzes

Test case ID	Requirement / feature	Test objective	Expected result	Status
TC-01	Login	Verify authentication	User logs in successfully	Pending
TC-02	Upload content	Verify upload	Content saved correctly	Pending
TC-03	Generate quiz	Verify AI output	Quiz generated for content	Pending
TC-04	Submit quiz	Verify scoring	Score calculated correctly	Pending
TC-05	Dashboard	Verify analytics	Data displayed correctly	Pending
TC-06	Offline Sync	Verify offline data storage and synchronization	Data is stored offline and synced when connection is restored	Pending
TC-AI-01	AI feature / response control	Verify the AI component produces acceptable output and hands over to a human or fallback process when confidence is low or rules are violated.	Output is relevant, safe, logged, and routed correctly according to defined acceptance criteria.	Pending
TC-AI-02	AI-05 — AI fallback	Verify fallback to manual quiz creation on LLM API timeout	After 30s timeout, system displays manual quiz editor; no error shown to learner	Pending

9.Dependability, Security, and Risk Management

This section addresses learning outcome H.1. The team must explain how the system and the development process will promote reliability, availability, safety, integrity, and recoverability. For AI-enabled projects, students should address dependability of models or external AI services, including reliability, privacy, monitoring, and safe degradation.

Dependability area	Planned technique / control	Why it matters for this project
Reliability	Use error handling and testing	Ensures system works consistently; error handling with retry (exponential backoff: 1s, 2s, 4s); circuit breaker on LLM API (5 failures → open 60s)
Availability	Cloud hosting (Vercel + Supabase)	System accessible anytime
Fault tolerance / recovery	Retry mechanisms + fallback (manual quiz)	Prevent system failure
Security / access control	Authentication + password hashing	Protects learner personal data (minors) per POPIA; JWT with 1-hour access + 7-day refresh tokens; bcrypt password hashing; AES-256 at rest; TLS 1.3 in transit; RBAC roles
Backup / logging / auditability	Database backups and logs	Prevent data loss
Operational risks	Internet dependency, API failure	Mitigate with caching and fallback
AI safety / model governance	Human review + logging AI outputs	Prevents incorrect or biased content from reaching learners; maintains teacher authority over educational content quality

10.Project Plan and Milestones

Provide a realistic work plan for the semester or project period. The plan should show who is responsible for what and when major artefacts will be produced.

Phase / sprint	Major activities	Expected deliverables	Responsible member(s)	Target date
1	Requirements & Design	ERD, proposal	All	Week 1–2
2	Backend Dev	API	Backend team	Week 3–4
3	Frontend Dev	UI	Frontend team	Week 5–6
4	AI Integration	Quiz generation	All	Week 7–8
5	Testing & Final	Report + demo	All	Week 9–10

11.Resources, Tools, and Feasibility

Resource category	Required item(s)	Availability / feasibility note
Hardware	Laptops, internet access	All team members have access to laptops and internet
Software / frameworks	FastAPI, Next.js, Supabase	All tools are open-source or free-tier and easily accessible online.

Cloud / hosting	Vercel, Supabase	Both platforms offer free tiers suitable for development and testing. Easy deployment and integration.
Data / datasets	Teacher-provided content	Data will be generated within the system by teachers, eliminating the need for external datasets.
Skills needed	Programming, AI integration, teamwork	Team members have academic background in Computer Science and will collaborate to cover all required skills.
AI tools / models / datasets	OpenAI API / LLM APIs	Accessible via API with free or low-cost tiers. Used for quiz generation. No need for training custom models.

12.Ethical, Legal, and Professional Considerations

- The system will handle **student personal data**, requiring secure storage and access control
- Users must **consent** to data collection and usage
- No sensitive data will be shared with AI services
- AI outputs will be reviewed to avoid **bias or incorrect content**
- Open-source tools and APIs will be used in compliance with licensing terms
- Transparency will be maintained by informing users when AI is used
- POPIA Compliance: All learner data encrypted (AES-256 at rest, TLS 1.3 in transit). JWT-based authentication with RBAC. No personal data shared with AI services.
- Informed Consent: Users must consent to data collection on registration. A clear privacy policy is displayed at sign-up.
- Child Safety: Learners are minors (ages 14–18). Row-level security ensures learners access only their own data. No social features expose learner identity to other users.
- Open-Source Compliance: All frameworks used under MIT or Apache 2.0 licences. OpenAI and Groq API usage complies with their terms of service for educational applications.

13.Expected Outcomes and Evaluation

Area	Student response
Expected final product	A working web-based AI-powered academic monitoring system with quiz generation and performance tracking.
Success indicators	• Users can register and log in • AI generates relevant quiz questions • Learners complete quizzes successfully • Teachers can track performance
Evaluation method	• System testing results • User feedback • Performance metrics (accuracy, response time)
Potential limitations	• AI may generate incorrect questions • Internet dependency • Limited time for advanced features
AI evaluation and user trust (if applicable)	• Teachers review 100% of AI outputs before publication to learners • Accuracy and relevance measured via UAT rating forms (teacher rating ≥4/5 for 85% of questions) • System logs ensure full transparency and audit trail; AI label displayed to learners on all AI-generated content

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Lecturer Assessment Guide

The rubric below may be used to evaluate the quality, realism, and academic completeness of the proposed project. Lecturers may adapt the weighting to suit module policy. In AI-enabled projects, lecturers should assess whether AI adds real engineering value and whether risks, limitations, and validation strategies are addressed credibly.

Criterion	Description	Suggested marks
Project relevance and originality	Clarity of problem, motivation, suitability for advanced diploma level, and originality of idea.	10
Requirements analysis	Quality of stakeholder analysis, functional and non-functional requirements, prioritisation, and specification.	10
Agile, DevOps, and process planning	Appropriate use of Agile framework, iteration planning, DevOps pipeline thinking, and process improvement strategy.	10
Architecture and design decisions	Appropriateness of architecture, distributed/service orientation considerations, and design pattern use.	10
Quality assurance and testing strategy	Definition of metrics, QA activities, test case planning, and acceptance criteria.	10
Dependability, reuse, and emerging trends	Practical treatment of reliability, reuse strategy, and justified modern software engineering trends.	10
Feasibility and planning	Realistic scope, team role allocation, resources, milestones, and risk awareness.	5

Presentation and professionalism	Structure, completeness, clarity, formatting, and quality of writing.	5
AI integration and responsible use	Clear justification for AI use or non-use, suitable AI method/tool selection, realistic data plan, measurable evaluation, and responsible AI controls.	10